EGA: Manifold-Preserving Adapter Training for Out-of-Distribution Retrieval in Vector Search

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Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through lightweight adapters, global objectives produce dense gradient fields that systematically distort the latent manifold of unseen categories, causing out-of-distribution (OOD) retrieval performance to collapse below the frozen baseline. To address this, we propose **Euclidean Geodesic Alignment (EGA)**, a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient regime in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space—including regions occupied by unseen categories—structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings. On strictly unseen fine-grained categories, competing adapters degrade below the frozen CLIP baseline in label precision, whereas EGA improves it by up to 18.5%.

1 Introduction

Contributions This paper makes the following contributions:

- **Phenomenon.** We identify a fundamental in-distribution/OOD tradeoff in global contrastive adapter training: while global objectives achieve higher geometric compression on seen categories, they produce dense gradient fields that systematically distort the pretrained manifold of unseen categories, causing retrieval performance to fall below the frozen CLIP baseline on out-of-distribution data.
- **Method.** We propose EGA, a lightweight residual adapter combining zero initialization, ℓ_2 normalization, and a small-margin triplet loss. By restricting optimization to locally violated neighborhood constraints, EGA refines the pretrained embedding space without destroying its global semantic structure, simultaneously improving retrieval quality on seen categories and preserving manifold integrity for unseen ones.
- **Mechanism and evaluation.** We provide a mechanistic explanation grounded in gradient sparsity: EGA's triplet objective converges to a regime where 96.5% of training triplets contribute zero gradient, leaving unseen class regions structurally unchanged from CLIP. We validate this mechanism empirically and demonstrate through evaluations on three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101) that EGA outperforms ICon and SRL

Preprint.

on both label precision and ANNS recall, while competing methods degrade below the frozen baseline.

2 Related Work

Vision Language Representations and Semantic Manifolds Vision language foundational models, such as CLIP [14], have established a dominant paradigm for constructing joint semantic spaces through large scale contrastive learning. Despite their remarkable zero shot capabilities, recent studies reveal that these continuous representations suffer from inherent geometric failures [17]. Specifically, the modality gap and cross modal misalignment lead to feature spaces that are overly sparse and structurally fractured [15, 3]. Furthermore, attempts to align complex compositional concepts often struggle in standard Euclidean spaces, highlighting the systemic disconnect between semantic meaning and spatial proximity [13]. This geometric illusion is exacerbated by the representation degradation problem prevalent in contrastive objectives. Extensive research indicates that latent embeddings tend to occupy narrow conic regions, a phenomenon known as anisotropy, which renders standard similarity metrics ineffective [27]. When models undergo continual learning or adapt to imbalanced fine grained categories, the feature space becomes increasingly distorted [11, 23]. Consequently, the dense gradient updates required by conventional multi view alignment often destruct specific latent structures rather than preserving them [22]. To mitigate these issues, recent adaptation strategies impose strict geometric constraints, such as orthogonality regularization, to balance new knowledge acquisition with the stability of the original representation [16]. While these methods attempt to preserve the global topological structure, they still rely on global optimization pressures that inadvertently alter the continuous manifold. Unlike these approaches, EGA employs a lightweight residual adapter with local geometric anchors to smooth specific class clusters, thereby correcting the Euclidean distance metric without subjecting the broader pre trained semantic relations to destructive global gradients.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [19, 10]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution, a strategy applied across various domains including graph encoders [20] and multimodal recommendation systems [28]. While pushing embeddings toward a uniform distribution enhances robustness to background noise [21], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like I Con formulate a unifying framework to systematically align multi view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [2]. Other approaches attempt to decouple the learning objectives [9] or apply information theoretic regularizers [26] to control the density of the learned representations. Furthermore, region based distillation techniques have been proposed to enhance fine grained understanding [7]. However, these global contrastive objectives and structural regularizers typically generate dense gradient fields throughout the training process. By continuously applying optimization pressure across the entire batch, they inadvertently disrupt the delicate pre established geometric structures of unseen fine grained categories. Unlike these global approaches that aggressively reconfigure the entire latent manifold through dense updates, EGA utilizes local geometric anchors to perform sparse and targeted metric smoothing, thereby preserving the structural integrity of out of distribution semantics.

Parameter Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [4, 5]. This paradigm has extensively shifted towards vision language models, dominating recent literature due to the prohibitive cost of full scale fine tuning [12]. Methods such as Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [25]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine tuning while preserving the generalizable representations of the foundation model [18]. To further protect the pre trained manifold during initial training steps, zero initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [24]. While these existing methods successfully maintain global knowledge and prevent catastrophic forgetting, they are not designed to resolve local metric inconsistencies for vector search. Unlike these general adaptation strategies, EGA integrates a zero initialized residual adapter with a localized geometric objective, explicitly smoothing class specific clusters to correct Euclidean distances without compromising the parameter efficiency or stability of the pre trained manifold.

Vector Similarity Search and Metric Consistency

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \to \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

Label Precision@K measures the semantic quality of retrieved results—whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\left[y_{\hat{n}_k(q)} = y_q\right], \tag{1}$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k-th nearest neighbor of q in the transformed embedding space.

ANNS Recall@K measures the geometric fidelity of the ANN index—the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget nprobe:

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{\left|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)\right|}{K}, \tag{2}$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top-K neighbor sets, respectively.

The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@K by making the geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4, global contrastive objectives achieve high AR@K on seen classes but collapse on *both* metrics for unseen classes, whereas EGA improves both simultaneously under the OOD setting.

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is deliberately constrained by three design principles, each of which serves a distinct role in preserving the pretrained manifold.

Residual connection with zero initialization. Each EGA block applies a residual update of the form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \tag{3}$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation

(Table 1), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

Feature gating. Inside each residual block, the transformed signal passes through a feature-gating module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2\, \text{GELU}(W_1 \mathbf{h}))\,, \tag{4}$$

where $W_1 \in \mathbb{R}^{(d/4)\times d}$, $W_2 \in \mathbb{R}^{d\times(d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise multiplication. This structure, analogous to Squeeze-and-Excitation networks [6], learns a per-dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio of $1/4$ is chosen to prevent the gating network from overfitting on seen-class data, as validated in Table 2.

ℓ_2 normalization. The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \tag{5}$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage points (Table 1), confirming that hypersphere consistency is necessary for stable ANN indexing.

Full forward pass. Let Block_i denote the i-th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm. The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z}))))\,. \tag{6}$$

The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension 2048, which is negligible compared to the frozen CLIP backbone ($\sim$150M parameters).

3.3 Local Triplet Training and Gradient Sparsity

Training objective. Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a, and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n)\in\mathcal{B}} \max(0,\ d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m)\,, \tag{7}$$

where $d(\cdot,\cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis in Table 2.

Gradient sparsity and manifold preservation. A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \tag{8}$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 8 (left) shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast majority of local neighborhood relationships in the embedding space.

This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter's updates remain local—i.e., that gradients from seen-class triplets do not propagate into unseen-class regions of the embedding space—the CLIP manifold structure for unseen classes is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

Contrast with global contrastive objectives. Global objectives such as ICon [1] and SRL [2] compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a KL divergence between the empirical similarity distribution and the class-membership distribution, while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives generate dense gradient fields in every training step: every sample pair—regardless of whether its local geometry already satisfies the retrieval objective— contributes to the loss and receives a non-zero gradient. This global optimization pressure from seen-class training propagates throughout the embedding space, distorting the local manifold structure that CLIP had established for unseen categories.

Proposition 1 (Gradient Sparsity at Convergence)**.** *Let f_θ be trained with the triplet loss in Eq. (7) with margin $m > 0$ on a dataset $\mathcal{D}_{train}$ drawn from seen classes $\mathcal{Y}_{seen}$. Define the* active triplet ratio *at training step t as*

$$\rho(t) = \Pr_{(a,p,n)\sim\mathcal{B}}\left[d\left(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}\right) - d\left(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}\right) + m > 0\right]. \tag{9}$$

As f_θ converges, $\rho(t) \to \rho^*$, where

$$\rho^* = \Pr_{(a,p,n)}\left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0\right] \tag{10}$$

is determined solely by the margin m and the within-class distance distribution of the converged embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.

Proof Sketch. The loss in Eq. (7) is a sum of hinge terms, each of which has gradient zero whenever $d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m \leq 0$. At convergence, f_θ has minimized the loss over $\mathcal{D}_{\text{train}}$, so the only remaining active triplets are those whose geometry the adapter has not yet been able to separate within the margin. The fraction of such triplets is bounded below by the overlap of the within-class and between-class distance distributions at the converged embedding, which depends on m and the geometry of $\mathcal{Y}_{\text{seen}}$ but not on $\mathcal{Y}_{\text{unseen}}$. Since no triplet involving unseen-class embeddings is ever constructed, the adapter gradient is identically zero over the entire unseen-class region of the embedding space throughout training. See Appendix A for the complete proof. □

The empirical validation of Proposition 1 is provided in Section 4.7, where we track the active triplet ratio throughout training and connect it directly to the OOD retrieval outcomes shown in Figure 8.

4 Evaluation

4.1 Experimental Setup

Experiment Platform All experiments were conducted on the Chameleon Cloud [8] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS. [Dongfang: Provide more system setup info.]

4.2 Ground truth of label precision

The metric of label precision is employed to assess the semantic quality of the retrieved results within the approximate nearest neighbor framework. This evaluation focuses on whether the neighbors found in the feature space are categorized under the same physical class as the query. By examining different neighborhood sizes from K=1 to K=10, we can determine the local and global semantic density of the manifold. Our analysis contrasts the original latent manifold with the EGA flattened model to highlight the impact of manifold smoothing on search accuracy.

Figure 1 demonstrates that EGA flattened achieves a substantial improvement in label precision across all configurations. In the K=1 scenario, our method maintains a precision score exceeding 0.7, while the original latent manifold fluctuates between 0.5 and 0.6. This performance gap remains persistent as the search budget nprobe increases from 1 to 10, indicating that the benefits of our approach are not dependent on extensive search efforts. The superior semantic alignment shown in

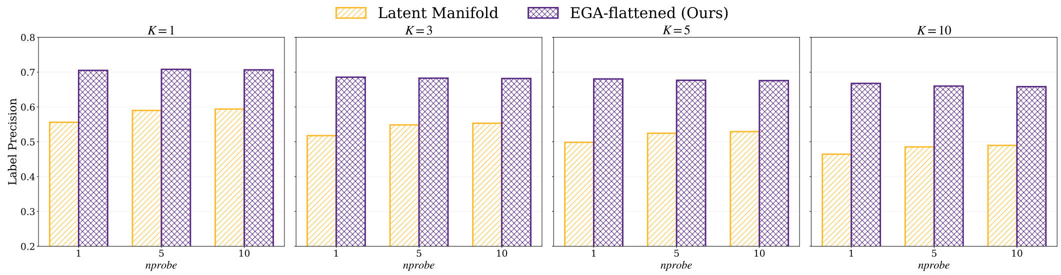

Figure 1: Label precision across varying K and nprobe settings. EGA consistently outperforms the original latent manifold, demonstrating improved semantic alignment.

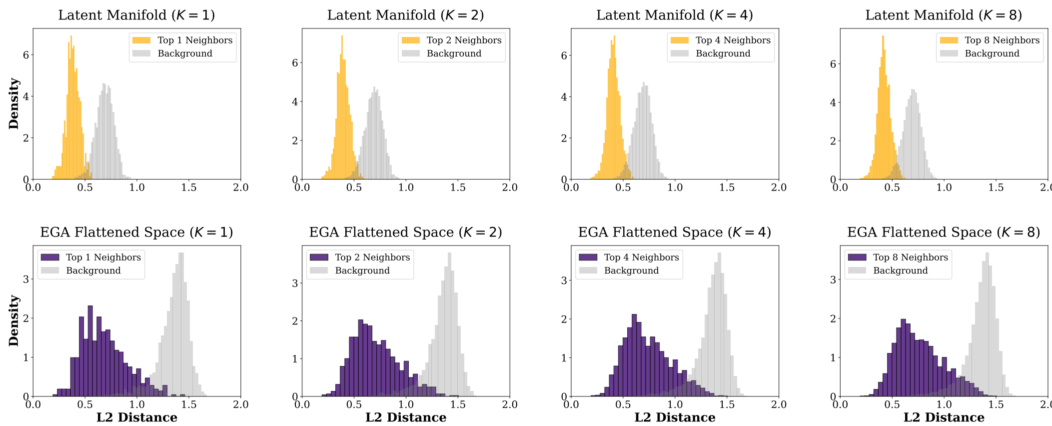

Figure 2: L2 distance distributions for Top K neighbors versus random background points. The original space exhibits significant overlap, whereas EGA enforces a clear geometric margin.

these figures suggests that EGA creates a more organized and discriminative feature space, allowing the indexer to retrieve correct class labels with higher reliability than the baseline.

4.3 Distances in Latent and EGA Spaces

To validate our stance on the systemic disconnect between semantic alignment and geometric proximity, we present empirical evidence across two dimensions: physical distance distributions and retrieval efficiency metrics. We evaluate Euclidean Geodesic Alignment (EGA) against raw foundational model embeddings using CIFAR10 and CIFAR100 datasets.

The results in Figure 2 expose the root cause of retrieval failure. In the original CLIP latent manifold, the distance distributions of true semantic neighbors and random background points exhibit significant overlap. **This position paper argues that this overlap constitutes a geometric illusion where noise is indistinguishable from signal, rendering Euclidean indexing inherently unreliable.**

4.4 Recalls in Latent and EGA Spaces

The impact of this flattening is quantified in Figure 3. The raw CLIP space achieves a meager 0.51 Recall@1 at nprobe=1 on CIFAR100. To compensate for this disorganized geometry, systems are forced to increase search volume to nprobe=10 to reach a recall of 0.85. This represents a 10x waste of computational resources.

In contrast, EGA elevates the nprobe=1 Recall@1 to 0.68. Notably, EGA at nprobe=1 outperforms the original space at nprobe=5 across all benchmarks. This confirms that fixing the geometry at the source is the only viable path to scalable retrieval. We achieve accuracy parity with brute force multi probe baselines while maintaining the extreme efficiency of single cell lookups. These results verify that our position is a necessary requirement for the next generation of scalable retrieval systems.

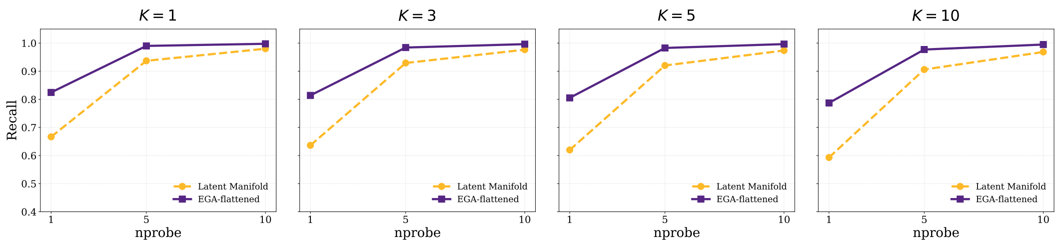

Figure 3: Recall@K versus nprobe tradeoff. EGA consistently shifts the efficiency frontier upward, enabling high recall at minimal search probes.

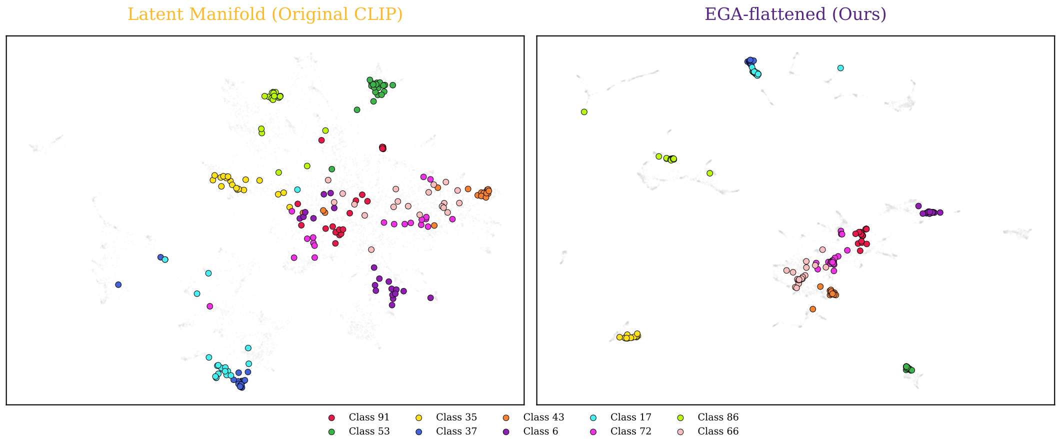

Figure 4: tSNE projection of original CLIP latent manifold (left) versus EGA flattened manifold (right). EGA reveals clearer category clusters and reduced semantic overlap.

4.5 Low dimensional projection visualization

To better understand the structural changes in the feature space, we utilize t SNE to project high dimensional embeddings into a two dimensional plane. Figure 4 compares the spatial distribution of the latent manifold generated by original CLIP against our proposed method. As observed in the left panel, the baseline manifold is characterized by highly dispersed data points and substantial semantic overlap between different categories. For instance, classes such as Class 91 and Class 53 appear entangled in certain regions, which inherently limits the effectiveness of search algorithms.

The right panel of the figure displays the EGA flattened manifold, revealing a significant enhancement in geometric order. Data points corresponding to specific categories, including Class 35 and Class 72, are tightly clustered and separated by well defined boundaries. This visual evidence confirms that our smoothing technique effectively eliminates manifold irregularities and reduces local noise. Such a structured feature distribution is a direct consequence of the manifold flattening process, which facilitates faster convergence in indexing and contributes to the superior recall rates discussed in previous sections.

4.6 Out-of-Distribution Generalization

The generalization capabilities of EGA are evaluated on the CIFAR-10 dataset, providing a rigorous out-of-distribution (OOD) test since all models were trained exclusively on CIFAR-100. Figure 5 illustrates the recall performance across varying nprobe values and K settings. EGA consistently maintains a dominant lead over all baselines, including recent state-of-the-art methods like ICon [1] and SRL [2].

At a minimum search budget of nprobe=1, EGA achieves a Recall@1 of 0.858, significantly outperforming SRL (0.831) and ICon (0.788). As shown in the inset zooms, EGA continues to hold a clear advantage even at nprobe=10, where other models begin to saturate. A key observation is

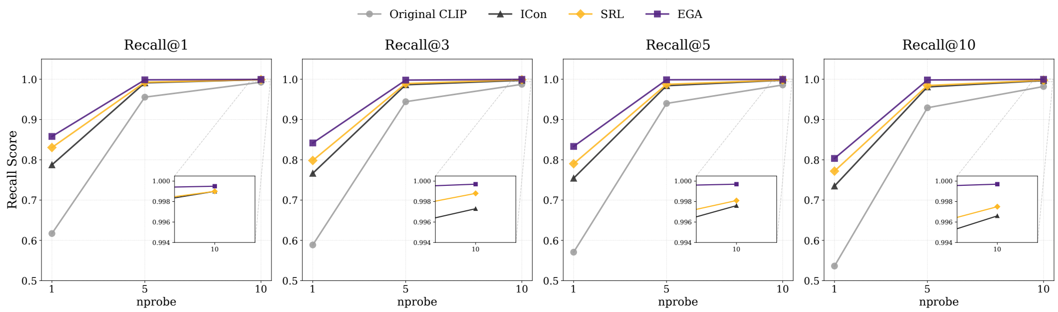

Figure 5: Recall@K versus nprobe tradeoff on CIFAR-10. EGA maintains a consistent lead over baselines, demonstrating strong OOD generalization.

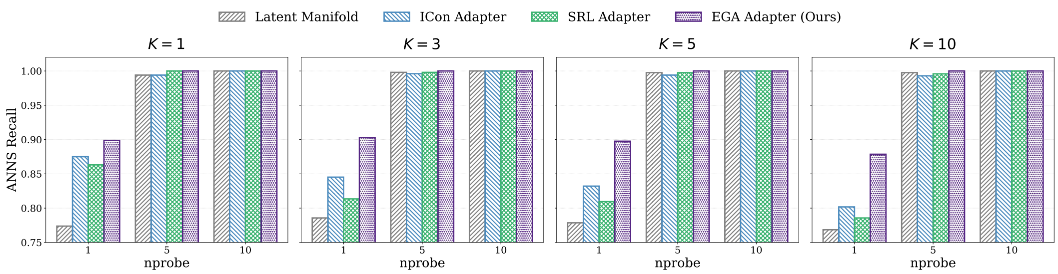

Figure 6: Recall versus nprobe tradeoff on 20 unseen FGVC Aircraft classes.

the efficiency gain provided by our method: EGA achieves a recall level at nprobe=5 that matches or exceeds the performance of supervised baselines at nprobe=10. This indicates that the manifold smoothing provided by EGA effectively doubles the search efficiency in unseen data environments by creating a more index-friendly geometric structure.

To further validate the robustness of EGA under fine-grained domain shifts, we extend our evaluation to the FGVC Aircraft dataset. Figure 6 visually corroborates that EGA remains the only method capable of scaling zero-shot retrieval performance on fine-grained domains without destroying the underlying manifold. In this setting, models are trained on 80 known aircraft categories and tested on 20 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial preservation capabilities. Global contrastive and structural regularization methods, such as ICon and SRL, suffer severe performance degradation when confined to a lightweight adapter bottleneck. Specifically, at $K = 1$ and nprobe=1, the recall for ICon and SRL collapses to 0.422 and 0.440 respectively, falling significantly below the original frozen CLIP baseline of 0.773. This phenomenon indicates that global contrastive objectives aggressively distort the latent manifold of unseen fine-grained categories when the representation capacity is constrained. Conversely, EGA exhibits exceptional stability and generalization, elevating the recall to 0.898 under the exact same adapter constraints. By leveraging local geometric anchors rather than forcing global distributions, EGA successfully compresses class-specific clusters while maintaining the structural integrity of the broader semantic space.

To comprehensively evaluate the versatility of EGA across diverse fine-grained domains, we extend our evaluation to the Food-101 dataset. Figure 7 visually corroborates that EGA remains the only method capable of scaling zero-shot retrieval performance on fine-grained domains without destroying the underlying manifold. In this setting, models are trained on 80 known food categories and tested on 21 strictly unseen categories. For a fair comparison, all baseline methods are restricted to training identical lightweight MLP adapters on top of frozen CLIP features, thereby eliminating the parameter advantage of end-to-end backbone fine-tuning. The empirical results reveal a stark contrast in spatial preservation capabilities, particularly given the strong pre-trained representations of common food items. Global contrastive and structural regularization methods, such as ICon and SRL, suffer

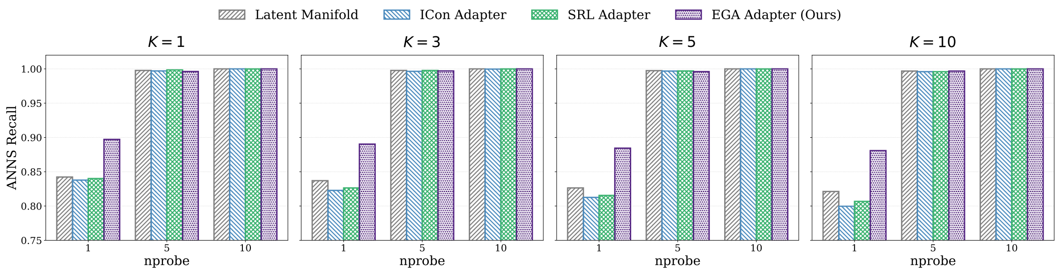

Figure 7: Recall versus nprobe tradeoff on 21 unseen Food-101 classes.

performance degradation when confined to a lightweight adapter bottleneck. Specifically, at $K = 1$ and nprobe=1, the recall for ICon and SRL drops to 0.838 and 0.840 respectively, falling below the original frozen CLIP baseline of 0.842. This phenomenon indicates that global contrastive objectives aggressively distort the already well-structured latent manifold of unseen categories when the representation capacity is constrained. Conversely, EGA exhibits exceptional stability and generalization, elevating the recall to 0.897 under the exact same adapter constraints. By leveraging local geometric anchors rather than forcing global distributions, EGA successfully compresses class-specific clusters while maintaining the structural integrity of the broader semantic space, significantly enhancing retrieval efficiency even on a highly optimized baseline manifold.

4.7 Mechanistic Analysis: Gradient Sparsity as OOD Protection

The superior OOD generalization of EGA raises a fundamental question: why does a triplet-trained adapter preserve unseen class structure while global contrastive objectives destroy it? We identify **gradient sparsity** as the key mechanism.

A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \tag{11}$$

As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 8 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by convergence. At steady state, **96.5% of triplets contribute zero gradient**, meaning the adapter updates only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space—including regions occupied by unseen classes—structurally intact.

Global contrastive objectives such as ICon [1] and SRL [2] exhibit the opposite behavior. ICon minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure from seen-class training propagates into unseen regions of the embedding space, distorting the local manifold structure that CLIP had established for those categories.

Figure 8 (right) quantifies the consequence on FGVC Aircraft, where all models are trained on 80 seen classes and evaluated strictly on 20 unseen classes. ICon and SRL collapse to label precision of 0.423 and 0.441, respectively—**below the frozen CLIP baseline of** 0.512—confirming that dense gradient updates actively harm zero-shot retrieval. EGA, by contrast, improves to 0.548 while modifying only 3.5% of the embedding geometry at convergence.

This analysis points to a key distinction in adapter design for foundation models: **objectives that produce dense gradient fields throughout training systematically degrade the pretrained manifold for unseen categories, while objectives that converge to sparse local updates preserve it**. The contrast is not merely one of degree—ICon and SRL fall below the frozen CLIP baseline entirely, suggesting that dense global optimization actively destructs zero-shot retrieval structure rather than simply failing to improve it.

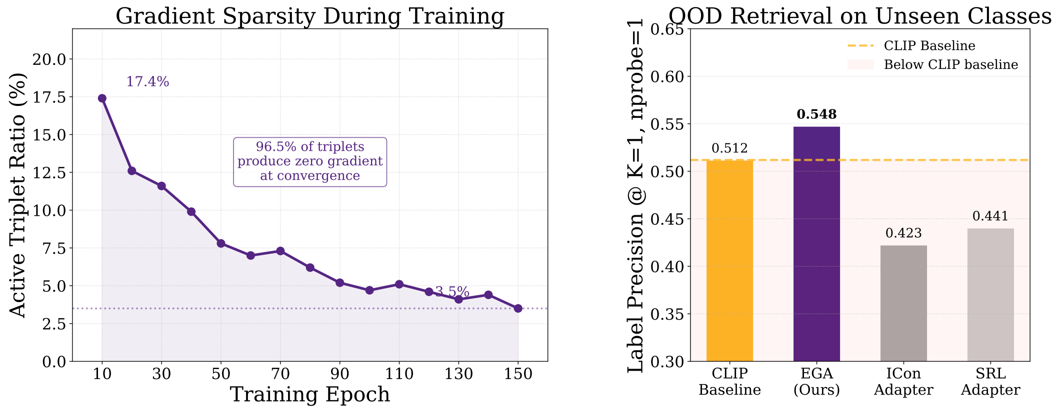

Figure 8: **Gradient sparsity as the mechanism behind EGA's OOD generalization.**

Table 1: Ablation study of EGA components on CIFAR-100.

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

4.8 Ablation Study

We perform a comprehensive ablation study on the CIFAR 100 dataset to verify the effectiveness of our design components. The results are summarized in Table 1. Each architectural choice contributes to the structural integrity of the calibrated manifold.

The residual connection is the most essential part of the design. When the residual connection is removed, the accuracy collapses from 0.7015 to 0.0065. This failure proves that the EGA adapter should refine the existing semantic space rather than replace it. Without this shortcut, the lightweight MLP cannot reconstruct the complex semantic relationships from limited training data.

L2 normalization is also necessary for stable search performance. Its removal leads to a performance loss of approximately 4.2 percent. This confirms that the hypersphere constraint is vital for consistent distance metrics in vector similarity search. Finally, the zero initialization strategy provides a 1.7 percent gain. By starting the adapter as an identity mapping, the model preserves pre trained knowledge at the beginning of the process, which provides a superior starting point for manifold smoothing.

4.9 Hyperparameter Sensitivity Analysis

The internal stability of EGA is evaluated by varying the triplet margin m and the hidden dimension d of the adapter. As presented in Table 2, our method demonstrates a consistent performance plateau across a broad range of settings.

The results for $K = 5$ indicate that EGA maintains high retrieval quality even when the margin m fluctuates. While the $K = 1$ metric shows sensitivity to larger margin values, the overall stability of the manifold is preserved. This trend suggests that a moderate margin setting, specifically between 0.1 and 0.3, is optimal for calibrating fine grained latent spaces. Furthermore, the performance remains nearly identical as the hidden dimension d scales from 512 to 4096. This observation confirms that the geometric correction provided by EGA is structural in nature and does not rely on massive parameter counts.

Table 2: Sensitivity of retrieval performance to different hyperparameter configurations on CIFAR 100.

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

5 Conclusion

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Checklist

A Proof of Proposition 1

Proof. We prove each claim in turn.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|}\sum_{(a,p,n)\in\mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_\theta \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_\theta \delta_{apn}, \tag{12}$$

which is exactly zero whenever $\delta_{apn} + m \le 0$. Therefore, $\nabla_\theta \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_θ and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Zero gradient over unseen regions. Let $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$ denote the region of the embedding space occupied by unseen-class images under f_θ. By construction, all triplets (a, p, n) in $\mathcal{D}_{\text{train}}$ satisfy $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$, so $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$ for all training triplets. The gradient $\nabla_\theta \delta_{apn}$ depends on f_θ only through its evaluation at $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$, which lie outside $\mathcal{R}_{\text{unseen}}$. Consequently, no training step produces a gradient that directly modifies f_θ in $\mathcal{R}_{\text{unseen}}$, and the CLIP geometry in that region is preserved throughout training.

(4) Dependence on m. The residual distance gap δ^*_{apn} at convergence is bounded by the within-class diameter $\Delta_c = \max_{x,x'\in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c. For $m \le \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^*. This explains the empirically observed increase in ρ^* with larger m reported in Table 2. □